

ADITI KUMARI

+919503038180 | aditikumarii8180@gmail.com | Pune, Maharashtra, India | [LinkedIn](#) | [GitHub](#)

SUMMARY

Final-year Computer Science Engineering student with hands-on experience in Python backend development, API development, automation, and data-driven applications. Skilled in building REST APIs, backend systems, and automation workflows using Python, Flask, and FastAPI. Familiar with software testing concepts, Pytest, Playwright, PostgreSQL, Git, and Agile SDLC methodologies. Strong problem-solving abilities with experience working on scalable real-world applications and research projects.

SKILLS

Programming Languages : Java, Python
Backend /APIs : Spring Boot, Flask (Web Framework), RESTful APIs, MVC Architecture, Spring Security
Testing / QA: Pytest (Learning), Playwright (Learning), API Testing , Manual Testing, Test Case Writing, Debugging
AI / Machine Learning : Vector Embeddings, NumPy, Scikit-learn, Retrieval-Augmented Generation (RAG)
Concepts & Analytics : Data Preprocessing, Data Visualization, Agile SDLC Methodology
Databases / Cloud : PostgreSQL, MongoDB, Supabase, AWS, Azure DevOps
Tools & Technologies: Git, Github, Docker, Postman
Generative AI / LLMs: Prompt Engineering, Retrieval Patterns
Automation: Browser Automation using Selenium
Languages: English, Hindi

WORK EXPERIENCE

Indian Space Research Organisation (ISRO) - Research Intern, Ahmedabad, India Jul 2025 - Sep 2025

- Contributed to the development of advanced algorithms for tracking Low Earth Orbit (LEO) satellites.
- Engineered precise position and velocity prediction models using orbital mechanics principles.
- Collaborated with senior scientists to optimize computational accuracy for real-time satellite tracking.

ElevenSkills-Data Science Intern, Tamil Nadu, India(Remote) Dec 2025 -Present

Dynamic Knowledge Expansion Chatbot | Python, FAISS, LLMs

- Engineered a RAG-based conversational AI system integrating vector embeddings and similarity search.
- Implemented automated knowledge ingestion pipeline to update vector database without model retraining.
- Optimized contextual retrieval to reduce hallucinations and enhance response relevance.
- Designed scalable architecture supporting continuous knowledge expansion.

EDUCATION

B.Tech (Computer Science Engineering), G.H Rasoni College of Engineering (CGPA: 8.18)
HSC, Kendriya Vidyalaya, Mumbai (CGPA:7.90)
SSC, Kendriya Vidyalaya ,Mumbai (CGPA: 7.40)

PROJECTS

LEO Satellite Tracking and Collision Alert System using TLE Data(Flask, CesiumJS,) [Link](#) Apr 2025 -Jul 2025

- Built a web app to track and visualize LEO satellites in real time using TLE data and CesiumJS.
- Implemented collision detection and orbit prediction using **RK4 + J2 perturbation model**.
- Integrated Supabase authentication and interactive dashboards for secure satellite tracking.
- Achieved approximately 98% short-term orbit prediction accuracy.

Online Food Ordering System (Spring Boot, MongoDB, Spring Security) [Link](#) Aug 2025 -Nov 2025

- Built a full-stack web application with role-based authentication (3 user roles) using Spring Security.
- Developed **15+ RESTful APIs** following MVC layered architecture for scalable backend design.
- Implemented CRUD operations and 5-stage real-time order workflow for efficient order management.
- Designed MongoDB schema with 3 optimized collections ensuring scalable NoSQL data handling.

RasoiBot - AI-Powered Recipe Recommendation System(Python, KNN, NumPy, OpenCV) [Link](#) Jan 2025 -Apr 2025

- Built a recipe recommendation system using KNN and cosine similarity based on available ingredients.
- Integrated a custom image classification module for food recognition without pretrained models.
- Achieved **94%** recommendation accuracy, improving user satisfaction.

CERTIFICATIONS

Introduction to Data Engineering, Coursera Apr 2025
AWS Educate Introduction to Generative AI, Amazon Web Services Training and Certification Mar 2026
AWS Educate Introduction to Cloud 101, Amazon Web Services Training and Certification Apr 2026

PUBLICATIONS

A TLE-Based LEO Satellite Orbital Trajectory Monitoring, Ephemeris Propagation, and Conjunction Alert System Nov '25

- Conducted research on LEO satellite tracking using TLE-based propagation, ephemeris prediction, and conjunction analysis.
- Developed a web platform with interactive 3D visualization using CesiumJS and a Python–Flask backend
- Built a preliminary collision-screening system to enhance space situational awareness.